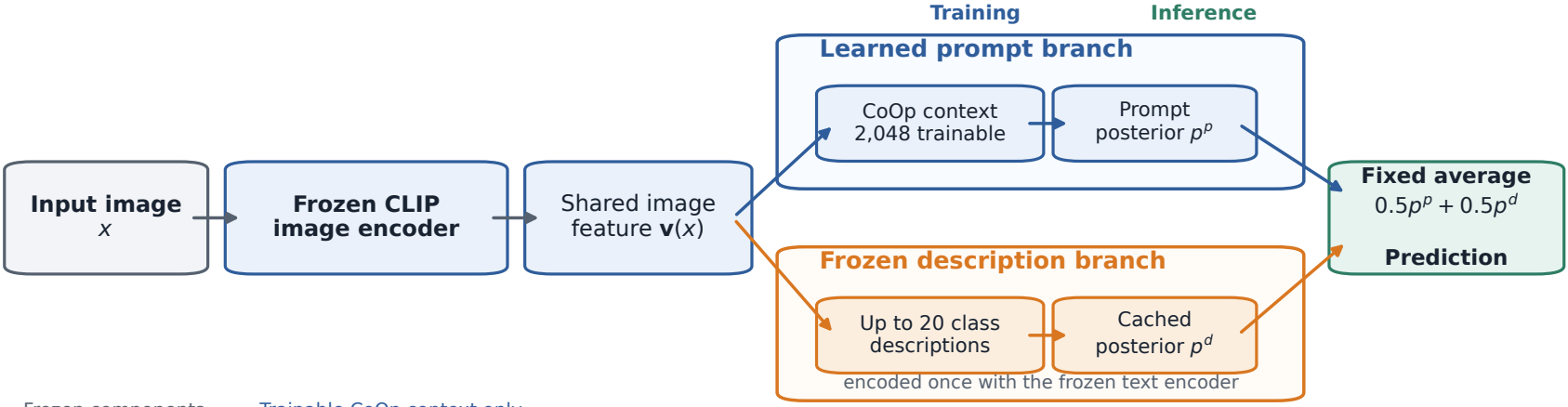
Training
Inference
Input image
x
Frozen CLIP
image encoder
Shared image
feature v(x)
Learned prompt branch
CoOp context
2,048 trainable
Prompt
posterior p^p
Frozen description branch
Up to 20 class
descriptions
Cached
posterior p^d
encoded once with the frozen text encoder
Fixed average
$0.5p^p + 0.5p^d$
Prediction
Frozen components
Trainable CoOp context only